

## Number System - Digital Circuits Questions and Answers - Sanfoundry

by Manish

This set of Digital Electronics/Circuits Multiple Choice Questions & Answers (MCQs) focuses on "Number System – 1".

1. Any negative number is recognized by its

- a) MSB
- b) LSB
- c) Bits
- d) Nibble

View Answer

Answer: a

Explanation: Any negative number is recognized by its MSB.

2. The parameter through which 16 distinct values can be represented is known as:

- a) Bit
- b) Byte
- c) Nibble
- d) Word

View Answer

Answer: c

Explanation: It can be represented up to 16 different values with the help of Nibble. Though, Nibble is a combination of four bits and it takes four bits to represent a single BCD or hexadecimal digit.

3. If the decimal number is a fraction then its binary equivalent is obtained by \_\_\_\_\_ the number continuously by 2.

- a) Dividing
- b) Multiplying
- c) Adding
- d) Subtracting

View Answer

Answer: b

Explanation: On multiplying the decimal number continuously by 2, the binary equivalent is obtained.

4. The representation of decimal number 532.86 in the form of decimal is

- a) 532.65
- b) 532.68
- c) 531.67
- d) 531.68

[View Answer](#)

Answer: b

Explanation:  $N = 5 * 10^2 + 3 * 10^1 + 2 * 10^0 + 8 * 10^{-1} + 6 * 10^{-2} = 532.68$

5. The binary equivalent of (1011.011)<sub>2</sub> is equal to

- a) 11.375
- b) 10.123
- c) 11.175
- d) 9.234

[View Answer](#)

Answer: a

Explanation:  $1 * 2^3 + 0 * 2^2 + 1 * 2^1 + 0 * 2^0 + 1 * 2^{-1} + 1 * 2^{-2} + 1 * 2^{-3} = 11.375$

Hence, (1011.011)<sub>2</sub> = 11.375

6. An important drawback of binary system is

- a) It requires very large string of 1's and 0's to represent a decimal number
- b) It requires sparingly small string of 1's and 0's to represent a decimal number
- c) It requires large string of 1's and small string of 0's to represent a decimal number
- d) None of the Mentioned

[View Answer](#)

Answer: a

Explanation: The most vital drawback of binary system is that it requires very large string of 1's and 0's to represent a decimal number.

7. The octal number 645 in power of 8 is equal to

- a) 450
- b) 451
- c) 421
- d) 501

[View Answer](#)

Answer: c

Explanation: The octal number 645 in power of 8 is given by:  $6 * 8^2 + 4 * 8^1 + 5 * 8^0 = 421$

8. The two digits hexadecimal number which has largest value is \_\_\_\_ which corresponds to

- a) FE, 255 decimal
- b) FF, 254 decimal

- c) FF, 255 decimal
- d) EF, 245 decimal

View Answer

Answer: c

Explanation: The largest of two digit hexadecimal number is FF and corresponds to  $15 * 16^1 + 15 * 16^0 = 255$  (since, F = 15)

9. Representation of hexadecimal number 6DE in the power of 16 is as:

- a)  $6 * 16^2 + 13 * 16^1 + 14 * 16^0$
- b)  $6 * 16^2 + 12 * 16^1 + 13 * 16^0$
- c)  $6 * 16^2 + 11 * 16^1 + 14 * 16^0$
- d)  $6 * 16^2 + 14 * 16^1 + 15 * 16^0$

View Answer

Answer: a

Explanation: In hexadecimal number D & E represents 13 & 14 respectively.

So,  $6DE = 6 * 16^2 + 13 * 16^1 + 14 * 16^0$ .

10. The quantity of double word is

- a) 16 bits
- b) 32 bits
- c) 64 bits
- d) 8 bits

View Answer

Answer: b

Explanation: The quantity of double word is 32 bits.

Digital Circuits Questions and Answers – Number System – 2

This set of Digital Electronics/Circuits Multiple Choice Questions & Answers (MCQs) focuses on "Number System – 2".

1. The given hex number (1E.53)<sub>16</sub> is equivalent to

- a) (35.68)<sub>8</sub>
- b) (35.24)<sub>8</sub>
- c) (34.34)<sub>8</sub>
- d) (35.59)<sub>8</sub>

View Answer

Answer: b

Explanation:  $(0001\ 1110.0101\ 0011)_2$

$= (00011110.01010011)_2$

$= (011110.010100)_2$

$= (011\ 110.010\ 100)_2$

$= (35.24)_8$

2. The octal number  $(651.124)_8$  is equivalent to

- a)  $(1A9.260)_{16}$
- b)  $(1B0.160)_{16}$
- c)  $(1A8.023)_{16}$
- d)  $(1B0.289)_{16}$

View Answer

Answer: a

Explanation:  $(651.124)_8 = (110\ 101\ 001.001\ 010\ 100)_2$   
 $= (110101001.001010100)_2$   
 $= (0001\ 1010\ 1001.0010\ 1010\ 0000)_2$   
 $= (1A9.260)_{16}$

3. The decimal equivalent of  $(LM17)_{16}$  is given by

- a) 491
- b) 499
- c) 497
- d) 498

View Answer

Answer: a

Explanation:  $(LM17)_{16} = 7 * 5^0 + 3 * 5^1 + M * 5^2 + L * 5^3$   
 $= 1 * 5^0 + 3 * 5^1 + 4 * 5^2 + 3 * 5^3$  ( $L=3, M=4$ )  
 $= 491.00$

4. Convert hexadecimal to binary:  $(1E2H) = ?$

- a) 480
- b) 483
- c) 482
- d) 484

View Answer

Answer: c

Explanation:  $(1E2)_{16} = 1 * 16^2 + E * 16^1 + 2 * 16^0 = 256 + 224 + 2 = 482$

5.  $(170)_{10}$  is equivalent to

- a)  $(FD)_{16}$
- b)  $(DF)_{16}$
- c)  $(AA)_{16}$
- d)  $(AF)_{16}$

View Answer

Answer: c

Explanation: Division Remainder

16) 170.00 ~~~~~—  
16) 10.00 ~~~~~10 = A  
16) 00.00 ~~~~~10 = A  
Hence,  $(170)_{10} = (AA)_{16}$

6. Convert in to decimal:  $(214)_8 = ?$

- a)  $(140)_{10}$
- b)  $(141)_{10}$
- c)  $(142)_{10}$
- d)  $(130)_{10}$

[View Answer](#)

Answer: a

Explanation:  $(214)_8 = 2 * 8^2 + 1 * 8^1 + 4 * 8^0$   
 $= 128.0 + 8.0 + 4 = (140)_{10}$

7. Convert  $(0.345)_{10}$  in to an octal number.

- a)  $(0.1605)_8$
- b)  $(0.2605)_8$
- c)  $(0.1945)_8$
- d)  $(0.2404)_8$

[View Answer](#)

Answer: b

Explanation:  $0.345 * 8 = 2.76 \ 2$

$0.760 * 8 = 6.08 \ 6$

$0.08 * 8 = 0.64 \ 0$

$0.640 * 8 = 5.12 \ 5$

$0.120 * 8 = 0.96 \ 0$

So,  $(0.345)_{10} = (0.2605)_8$

8. Convert from binary to decimal:  $(01011.1011)_2 = ?$

- a)  $(11.6875)_{10}$
- b)  $(11.5874)_{10}$
- c)  $(10.9876)_{10}$
- d)  $(10.7893)_{10}$

[View Answer](#)

Answer: a

Explanation:  $(01011)_2 = 0 * 2^4 + 1 * 2^3 + 0 * 2^2 + 1 * 2^1 + 1 * 2^0 = 11$

$(1011)_2 = 1 * 2^{-1} + 0 * 2^{-2} + 1 * 2^{-3} + 1 * 2^{-4} = 0.6875$

So,  $(01011.1011)_2 = (11.6875)_{10}$

9. Octal to binary conversion:  $(24)_8 = ?$

- a)  $(111101)_2$
- b)  $(010100)_2$
- c)  $(111100)_2$
- d)  $(101010)_2$

[View Answer](#)

Answer: c

Explanation:  $(24)_8 = (010100)_2$

10. Convert binary to octal:  $(110110001010)_2 = ?$

- a)  $(5512)_8$
- b)  $(6612)_8$
- c)  $(4532)_8$
- d)  $(6745)_8$

[View Answer](#)

Answer: b

Explanation:  $(110110001010)_2 = (6612)_8$

#### Digital Circuits Questions and Answers – Arithmetic Operation

This set of Digital Electronics/Circuits Multiple Choice Questions & Answers (MCQs) focuses on "Arithmetic Operation"

1. The addition of binary numbers:

$11011011010 + 010100101 = ?$

- a) 0111001000
- b) 1100110110
- c) 11101111111
- d) 10011010011

[View Answer](#)

Answer: c

Explanation: None.

2. Perform binary addition:  $101101 + 011011 = ?$

- a) 011010
- b) 1010100
- c) 101110
- d) 1001000

[View Answer](#)

Answer: d

Explanation: None.

2. Perform binary subtraction:  $101111 - 010101 = ?$

- a) 100100
- b) 010101
- c) 011010
- d) 011001

[View Answer](#)

Answer: c

Explanation: None.

3. The result obtained after  $(100101 - 011110)$  is

- a) 000111
- b) 111000

d) 101010

[View Answer](#)

Answer: a

Explanation: None.

4. Multiply the binary number:  $01001 \times 01011 = ?$

a) 001100011

b) 110011100

c) 010100110

d) 101010111

[View Answer](#)

Answer: a

Explanation: None.

5.  $100101 \times 0110 = ?$

a) 1011001111

b) 0100110011

c) 101111110

d) 0110100101

[View Answer](#)

Answer: c

Explanation: None.

6. On multiplication of (10.10) and (01.01), we get

a) 101.0010

b) 0010.101

c) 011.0010

d) 110.0011

[View Answer](#)

Answer: c

Explanation: None.

7. Divide the binary numbers:  $111101 \div 1001$  and find the remainder

a) 0010

b) 1010

c) 1100

d) 0011

[View Answer](#)

Answer: d

Explanation: None.

9. Divide:  $011010000 \div 0101 = ?$

a) 10001

- b) 10100
- c) 11001
- d) 01000

[View Answer](#)

Answer: b

Explanation: None.

10. Subtract:  $101101 - 001011 = ?$

- a) 100010
- b) 010110
- c) 110101
- d) 101100

[View Answer](#)

Answer: a

Explanation: None.

Digital Circuits Questions and Answers – 1's, 2's, 9's & 10's Complements – 1

This set of Digital Electronics/Circuits Multiple Choice Questions & Answers (MCQs) focuses on "1's, 2's, 9's & 10's Complements".

1. 1's complement of 1011101 is

- a) 0101110
- b) 1001101
- c) 0100010
- d) 1100101

[View Answer](#)

Answer: c

Explanation: None.

2. 2's complement of 11001011 is

- a) 01010111
- b) 11010100
- c) 00110101
- d) 11100010

[View Answer](#)

Answer: c

Explanation: None.

3. On subtracting  $(01010)_2$  from  $(11110)_2$  using 1's complements, the output will be

- a) 010010
- b) 110100
- c) 101011
- d) 110011

[View Answer](#)

Answer: d

Explanation: None.

4. On subtracting  $(010110)_2$  from  $(1011001)_2$  using 2's complements, we get

- a) 0111001
- b) 1100101
- c) 0110110
- d) 0101100

[View Answer](#)

Answer: d

Explanation: None.

5. On subtracting  $(001100)_2$  from  $(101001)_2$  using 2's complements, we get

- a) 1101100
- b) 011101
- c) 11010101
- d) 11010111

[View Answer](#)

Answer: b

Explanation: None.

6. On addition of 28 and 18 using 2's complements, we get

- a) 00101110
- b) 0101110
- c) 00101111
- d) 1001111

[View Answer](#)

Answer: b

Explanation: None.

7. On addition +38 and -20 using 2's complements, we get

- a) 11110001
- b) 100001110
- c) 00001110
- d) 110101011

[View Answer](#)

Answer: c

Explanation: None.

8. On addition -46 and +28 using 2's complements, we get

- a) 11101100
- b) 00010101
- c) 101011111
- d) 000100101

[View Answer](#)

Answer: a

Explanation: None.

9. The addition of -33 and -40 using 2's complements is equal to

- a) 01001110
- b) 111010101
- c) 10110001
- d) 10110001

[View Answer](#)

Answer: d

Explanation: None.

10. On subtracting +28 and +29 using 2's complements, we get

- a) 11111010
- b) 111111001
- c) 010101011
- d) 00000101

[View Answer](#)

Answer: d

Explanation: None.

## Digital Electronic Circuits Interview Questions and Answers - Sanfoundry

by Manish

This set of Digital Electronic Circuits Interview Questions and Answers focuses on “1's, 2's, 9's & 10's Complements-2”.

1. If the number of bits in the sum exceeds the number of bits in each added numbers, it results in

- a) Successor
- b) Overflow
- c) Underflow
- d) None of the Mentioned

[View Answer](#)

Answer: b

Explanation: If the number of bits in the sum exceeds the number of bits in each added numbers, it results in overflow and is also known as excess-one.

2. An overflow is a

- a) Software problem
- b) Hardware problem
- c) User input problem

1. Primary data for the research process be collected through \_\_\_\_\_.

- a. Experiment
- b. Survey
- c. Both a and b
- d. None of the above

Answer c

2. The data obtained by conducting a survey is called:

- a. Primary data
- b. Secondary data
- c. Continuous data
- d. Qualitative data

Answer: a

3. The data collected from published reports is known as:

- a. Discrete data
- b. Arrayed data
- c. Secondary data
- d. Primary data

Answer: c

4. A survey in which information is collected from each and every individual of the population is known as:

- a. Sample survey
- b. Pilot survey
- c. Biased survey
- d. Census survey

Answer: d

5. Data used by an agency which is originally collected them are:

- a. Primary data
- b. Raw data
- c. Secondary data
- d. Grouped data

Answer: a

6. Which form of data below can usually be obtained more quickly and at a lower cost than the others?

- a. Primary
- b. Survey research
- c. Experimental research
- d. Secondary

Answer: d

7. Secondary data are \_\_\_\_\_.

- a. Collected mostly via surveys
- b. Expensive to obtain
- c. Never purchased from outside suppliers
- d. Not always very usable

Answer: d

8. Secondary data consists of information \_\_\_\_\_.

- a. That already exists somewhere and was collected for another purpose
- b. Used by competitors
- c. That does not currently exist in an organised form
- d. That already exists somewhere and is outdated

Answer: a

9. Which method could a marketing researcher use to obtain information that people are unwilling or unable to provide?

- a. Focus groups
- b. Personal interviews
- c. Questionnaires
- d. Observational research

Answer: d

10. Survey research, though used to obtain many kinds of information in a variety of situations, is best suited for gathering \_\_\_\_\_ information.

- a. Attitudinal
- b. Personal
- c. Preference
- d. Descriptive

Answer: d

11. Survey research is least likely to be conducted through which of the following?

- a. Observation
- b. Person-to-person interactions
- c. The telephone
- d. The Web

Answer: a

12. In marketing research, the \_\_\_\_\_ phase is generally the most expensive and most subject to error.

- a. Interpreting and reporting the findings
- b. Data validation
- c. Data collection
- d. Planning

Answer: c

13. Marketing intelligence is everyday information about developments in the marketing environment that assists marketers in their preparation of their plans and strategies. This information is obtained from a number of sources and includes which of the following?

- a. Newspaper articles.
- b. Sales representative feedback.
- c. Competitor intelligence
- d. All of the above.

Answer: d

14. What are three popular methods for obtaining primary data?

- a. Experimentation; personal interview; Delphi technique.
- b. Survey; interviews; experimentation.
- c. Interviews and surveys; observation; experimentation.
- d. Interviews and surveys; observation; Harrison methodology.

Answer: c

15. Primary data is \_\_\_\_\_.

- a. Always collected before secondary data
- b. Collected for the specific purpose at hand
- c. Information that already exists
- d. Data collected for other purposes

Answer: b

16. For primary data to be useful to marketers, it must be relevant, current, unbiased, and \_\_\_\_\_.

- a. Complete
- b. Accurate
- c. Inexpensive
- d. Collected before secondary data

Answer: b

17. \_\_\_\_\_ method is the gathering of primary data by watching people.

- a. Survey
- b. Informative
- c. Observational
- d. Experimental

Answer: c

18. Interview is an example of \_\_\_\_\_ data

- a. Primary
- b. Secondary
- c. Both a and b
- d. None of the above

Answer: a

19. The existing company information is an example of which data?

- a. Primary
- b. Secondary
- c. Both a and b
- d. None of the above

Answer: b

20. Information collected from online databases is an example of \_\_\_\_\_ data.

- a. Primary
- b. Secondary
- c. Observational
- d. Experimental

Answer: b

21. Some of the advantages of secondary data are low cost, speed, availability, and flexibility.

- a. True
- b. False

Answer: a

22. To verify the quality of secondary data information, it may be necessary to know how the data were collected, what the sampling plan was, what data collection method was used, and what degree of nonresponse was experienced.

- a. True
- b. False

Answer: a

23. Providing the background necessary to understand the problem situation is a common use for

- a. primary data
- b. secondary data
- c. exploratory data
- d. explanatory data

Answer: b

24. If time is of the essence for a research project, the preferred contact method is:

- a. Mail
- b. Telephone
- c. Intercept interview
- d. Personal interview

Answer: b

25. If flexibility is the most important criterion for a research project, the preferred contact method is:

- a. Intercept interview
- b. Telephone
- c. Focus group
- d. Personal interview

Answer: d